INSTITUT UNIVERSITAIRE DES SCIENCES

Faculté des Sciences de Technologies

td-8 Réseaux 1

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Niveau : L3-Sciences Informatiques

Description :

→ Comprendre et apprendre à Configurez un pare-feu et un VPN site-à-site.

1. Configurez un pare-feu et un VPN site-à-site.

≫ *Configuration du Routeur, Switch*

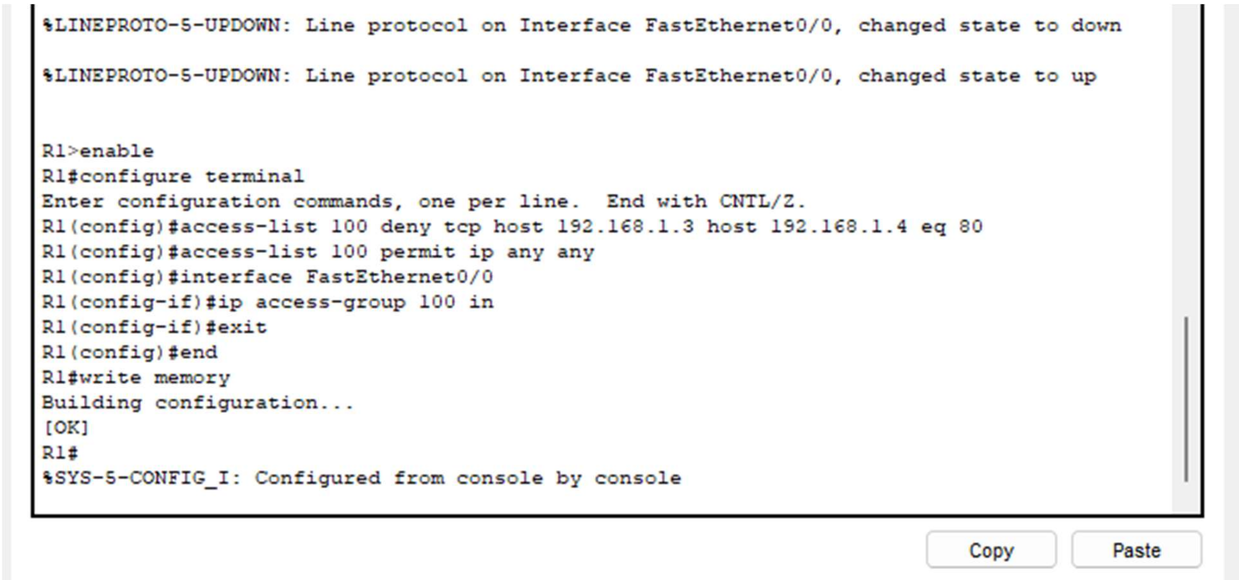
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config)#
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)## Configuration du nom du commutateur
                ^
% Invalid input detected at '^' marker.

Switch(config)#hostname MonCommutateur
MonCommutateur(config)#
MonCommutateur(config)## Configuration de l'interface VLAN 1
                        ^
% Invalid input detected at '^' marker.

MonCommutateur(config)#interface vlan 1
MonCommutateur(config-if)#description VLAN par dfaut
MonCommutateur(config-if)#ip address 192.168.1.2 255.255.255.0
MonCommutateur(config-if)#no shutdown

MonCommutateur(config-if)#
MonCommutateur(config-if)## Configuration d'un VLAN pour les utilisateurs
                           ^
% Invalid input detected at '^' marker.

MonCommutateur(config-if)#vlan 10
MonCommutateur(config-vlan)#name Utilisateurs
MonCommutateur(config-vlan)#
MonCommutateur(config-vlan)## Configuration d'un VLAN pour les voix
                             ^
% Invalid input detected at '^' marker.

MonCommutateur(config-vlan)#vlan 20
MonCommutateur(config-vlan)#name Voix
MonCommutateur(config-vlan)#
MonCommutateur(config-vlan)## Configuration des interfaces pour les VLANs
```

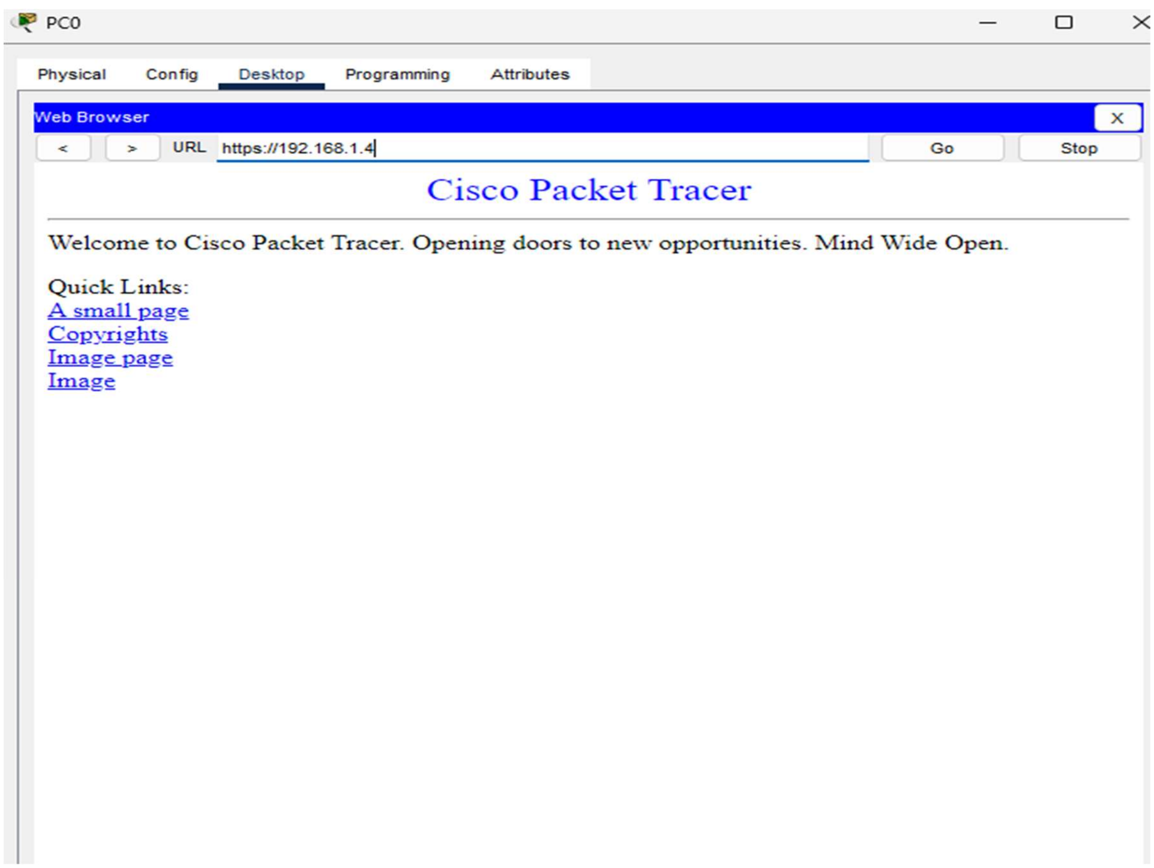
» vérifier la connectivite

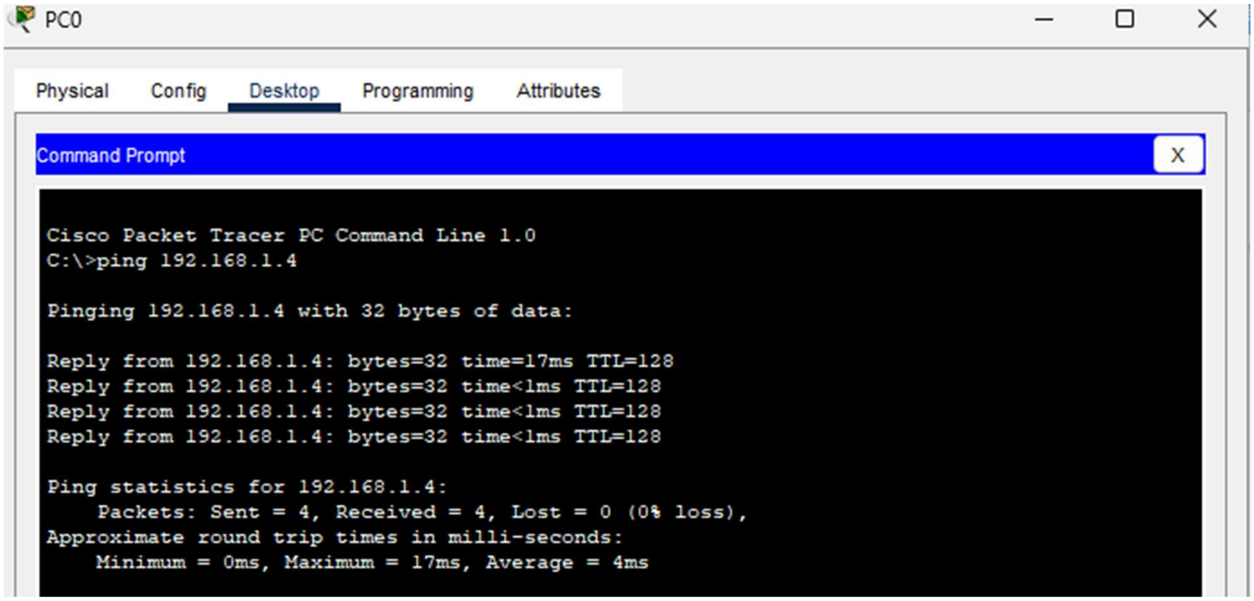
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:97FF:FEA5:AA2E
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.2
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

→> ACL

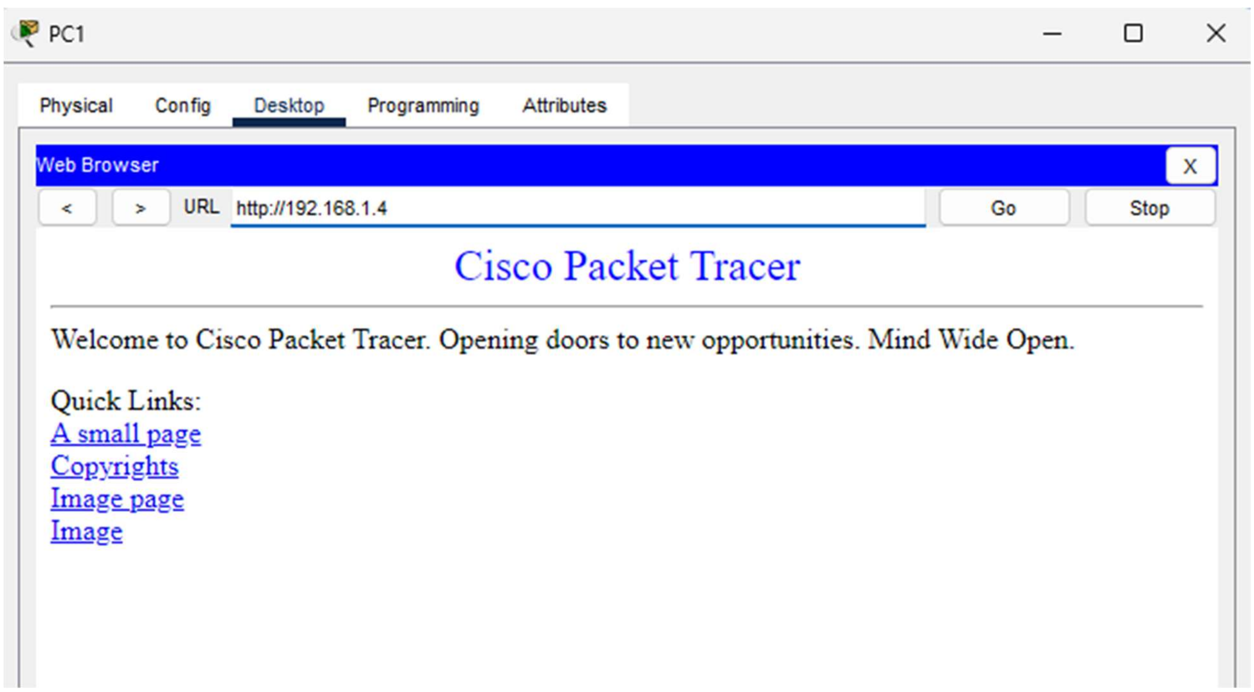


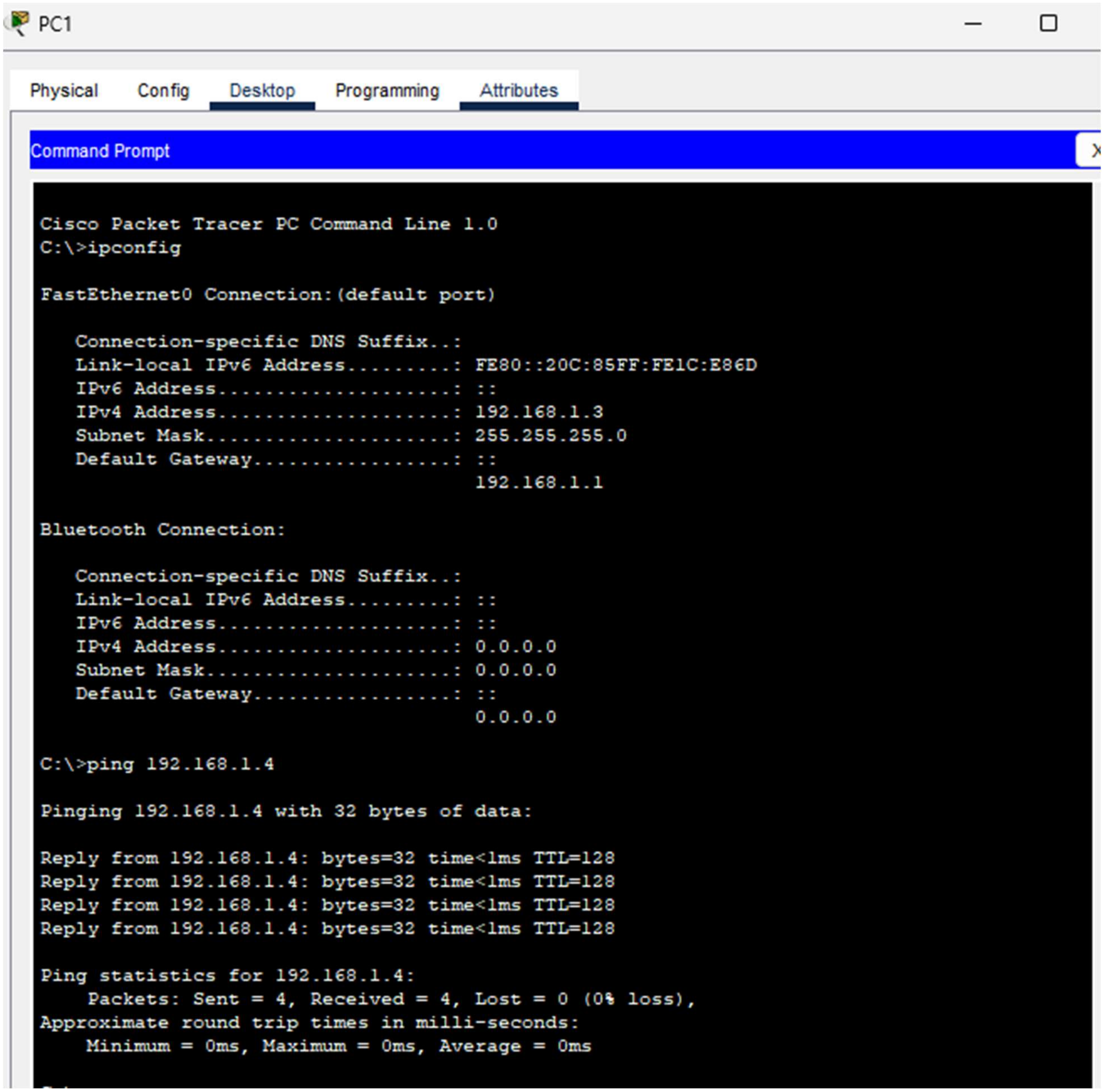
» Tester la Configuration




PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4
Pinging 192.168.1.4 with 32 bytes of data:
Reply from 192.168.1.4: bytes=32 time=17ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 17ms, Average = 4ms

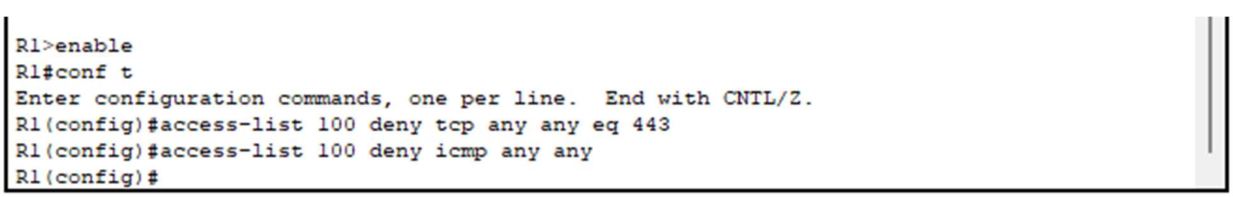
2eme PC


PC1
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://192.168.1.4
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::20C:85FF:FE1C:E86D
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.3
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

7. Bloquer le Trafic HTTPS et ICMP

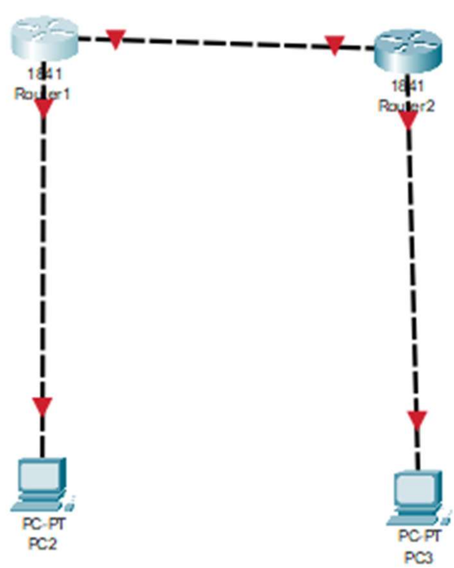


```
R1>enable
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 100 deny tcp any any eq 443
R1(config)#access-list 100 deny icmp any any
R1(config)#
```

```
R1(config)#interface FastEthernet0/0
R1(config-if)#ip access-group 100 in
R1(config-if)#exit
R1(config)#
```

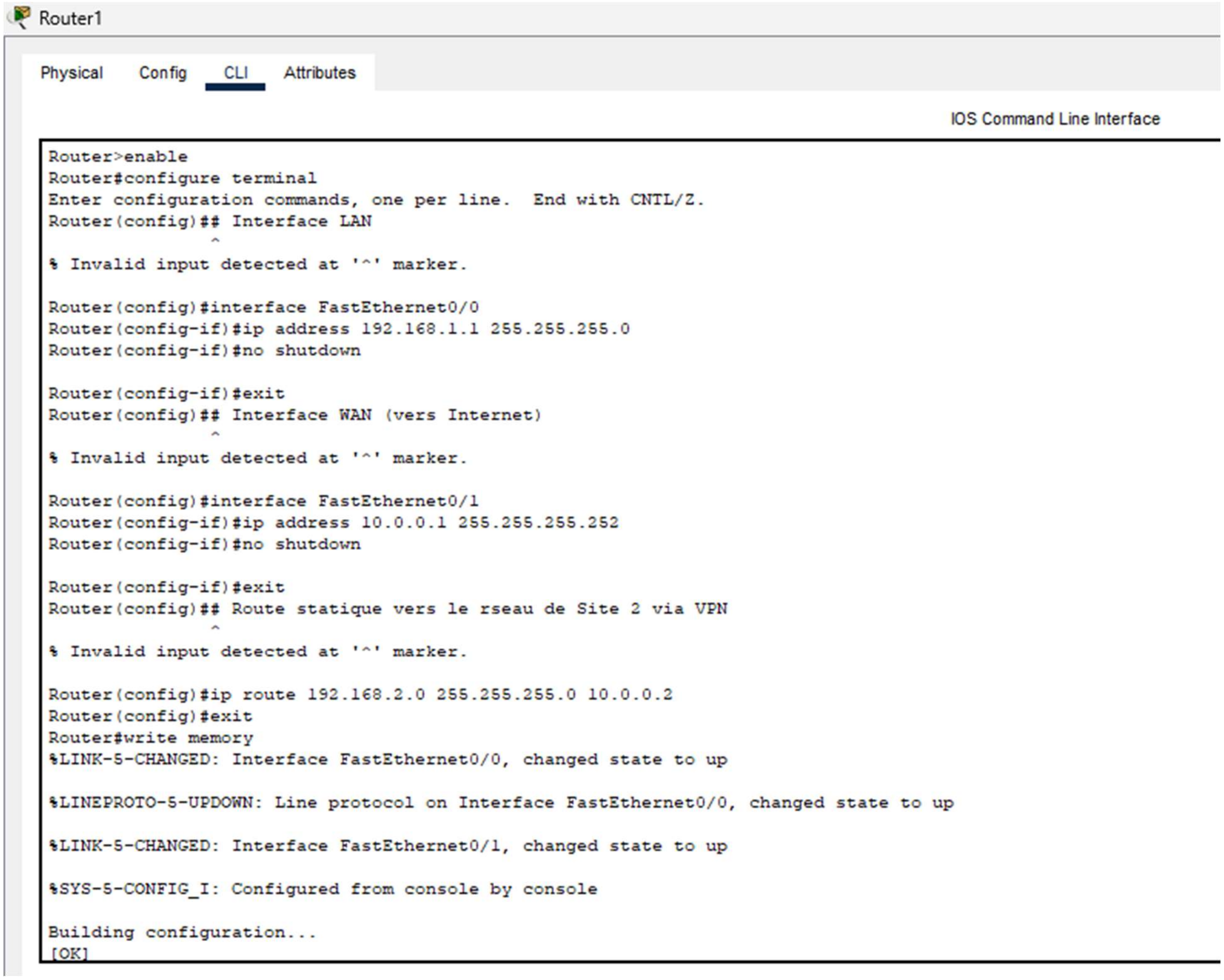
```
R1#write memory
Building configuration...
[OK]
R1#
```

Configuration d’un VPN Site-à-Site



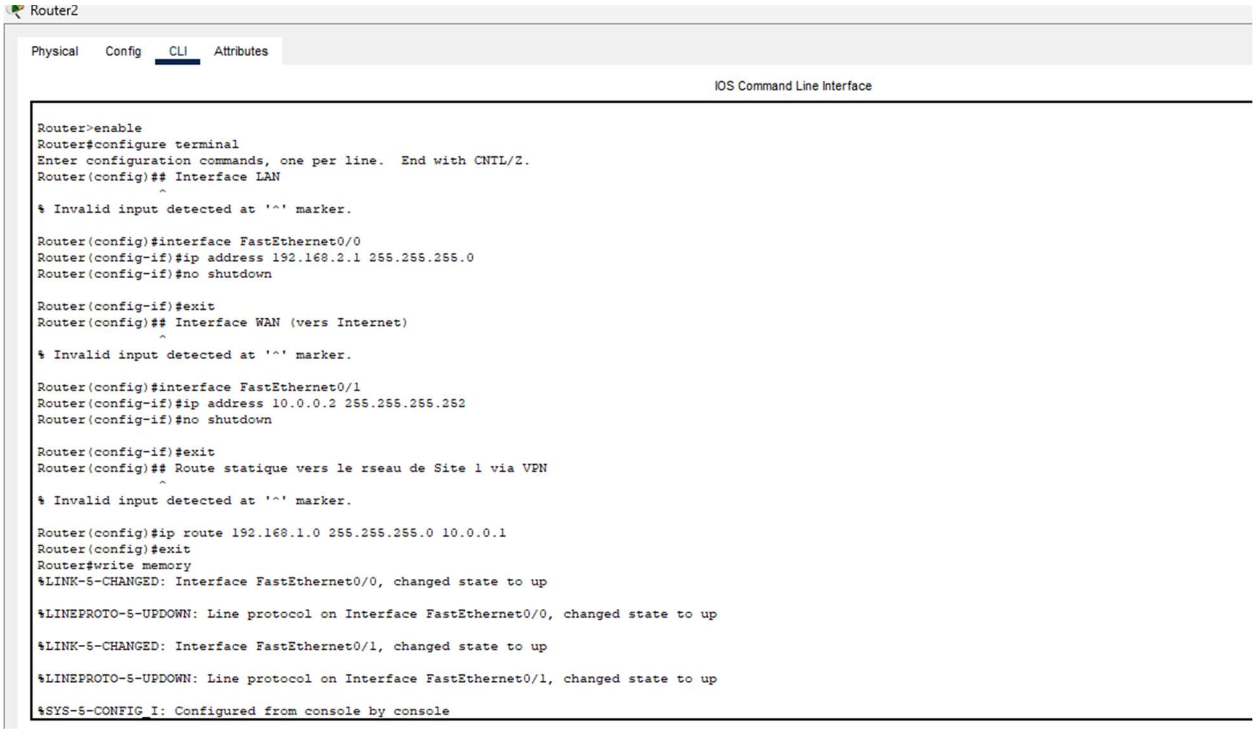
→ Configuration des Adresses IP

+> Routeur 1




```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)## Interface LAN
               ^
% Invalid input detected at '^' marker.

Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)## Interface WAN (vers Internet)
               ^
% Invalid input detected at '^' marker.

Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)## Route statique vers le rseau de Site 2 via VPN
               ^
% Invalid input detected at '^' marker.

Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#write memory
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

+> Routeur 2




```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)## Interface LAN
                ^
% Invalid input detected at '^' marker.

Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)## Interface WAN (vers Internet)
                ^
% Invalid input detected at '^' marker.

Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)## Route statique vers le rseau de Site 1 via VPN
                ^
% Invalid input detected at '^' marker.

Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#exit
Router#write memory
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
```

Configuration du VPN IPsec

+> R1

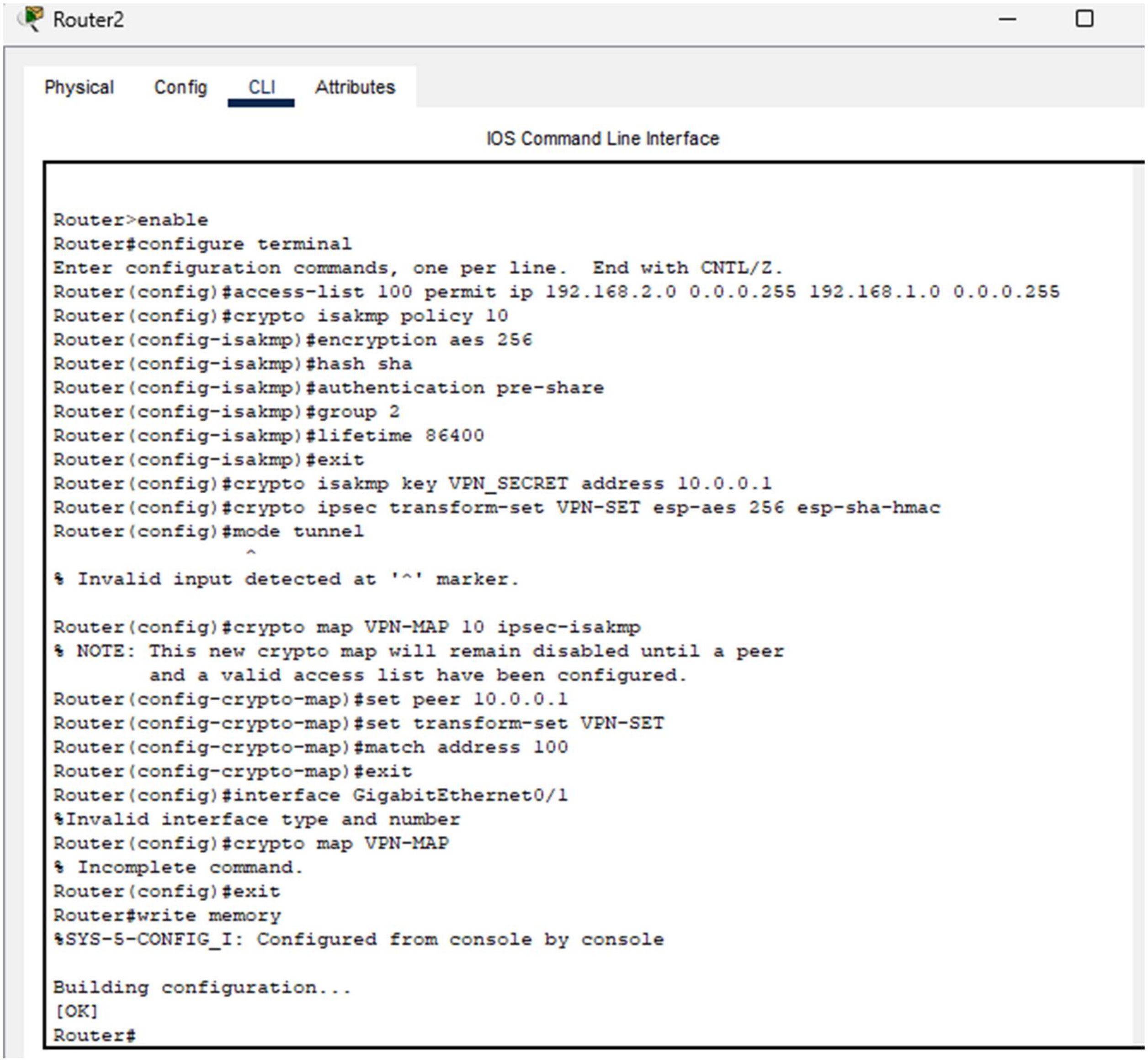
Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes 256
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key VPN_SECRET address 10.0.0.2
A pre-shared key for address mask 10.0.0.2 255.255.255.255 already exists!
Router(config)#crypto ipsec transform-set VPN-SET esp-aes 256 esp-sha-hmac
Router(config)#mode tunnel
                ^
% Invalid input detected at '^' marker.

Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.2
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface GigabitEthernet0/1
%Invalid interface type and number
Router(config)#crypto map VPN-MAP
% Incomplete command.
Router(config)#exit
Router#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

+> R2




```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes 256
Router(config-isakmp)#hash sha
Router(config-isakmp)#authentication pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key VPN_SECRET address 10.0.0.1
Router(config)#crypto ipsec transform-set VPN-SET esp-aes 256 esp-sha-hmac
Router(config)#mode tunnel
                ^
% Invalid input detected at '^' marker.

Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
Router(config-crypto-map)#set peer 10.0.0.1
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 100
Router(config-crypto-map)#exit
Router(config)#interface GigabitEthernet0/1
%Invalid interface type and number
Router(config)#crypto map VPN-MAP
% Incomplete command.
Router(config)#exit
Router#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

Vérification du VPN

→ Tester la connectivité

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Vérifier l'établissement du VPN

```
Router>enable
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status

IPv6 Crypto ISAKMP SA


Router#
```

```
Router#show crypto ipsec sa

No SAs found
```

Conclusion :

Ce TP m'a aidé à renforcer mes compétences en matière de sécurité réseau et à établir des connexions sécurisées entre des sites distants, tout en assurant une gestion efficace du trafic réseau. Ces compétences sont essentielles pour garantir la sécurité et l'efficacité des réseaux dans des environnements professionnels.